Declan Chan

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Education

BASc in Engineering Science *2025–Present*
University of Toronto
- Dean's Honours List (all semesters)

IB Diploma Programme *2021–2025*
Maple High School
- IB Score: 45/45

Technical Skills

Languages: Python, C, SQL, MATLAB, SystemVerilog, Bash

Libraries/Frameworks: PyTorch, NumPy, Matplotlib, ROS

Tools: Git, Linux, Slurm, SQLite, SolidWorks, OnShape

Experience

Software Developer *2025–Present*
Robotics for Space Exploration
- Developing a ROS 2 imitation-learning system to replace inverse kinematics for robotic manipulation.
- Building ROS bag workflows to collect joint-state data for transformer policy training.
- Evaluating LeRobot's ACT model for autonomous hammer pickup and equipment servicing.

Machine Learning Research Fellow *2026*
Autonomous Discovery of Alloys
- Evaluated the robustness of machine learning force fields (MLFFs) for trustworthy AI in materials discovery.
- Automated computational experiments using Python, Bash, and Linux on Canada's Fir HPC cluster.
- Benchmarked MLFFs against density functional theory in collaboration with postdoctoral researchers.
- Presented findings at Undergraduate Engineering Research Day Conference.

Space Systems Engineer *2026*
University of Toronto Aerospace Team
- Developed Python software for CubeSat attitude determination and control algorithm testing.
- Developed modular Python tools for attitude dynamics and flight simulation.
- Simulated magnetic environments using a Helmholtz cage and MATLAB.

Senior Instructor *2021–Present*
Northern Karate Schools
- Yondan: Certified 4th-degree Black Belt.
- Gold medal at the Toronto International Tournament of Martial Arts Champions.
- Led group and private karate classes for 100+ students each week.

Projects

Machine Learning Force Field Attacks *2026*
- Developed open-source software to generate gradient-based adversarial atomic perturbations.
- Built data-processing pipelines for Materials Project and Li-C-O datasets with GitHub and Hugging Face.
- Performed ASE relaxation and perturbations workflows using Bash and Slurm.
- Analyzed and visualized attack performance using PyTorch, NumPy, and Matplotlib.

Configurable PyTorch Deep Learning Lab *2026*
- Developed a configurable PyTorch image-classification pipeline supporting reproducible training.
- Designed modular training workflows supporting multiple datasets and model configurations.
- Automated experiment tracking with SQLite and SQL to compare accuracy of model configurations.

Medical RAG Platform *2026*
- Built an interactive 3D web application for visualizing human disease progression.
- Implemented a retrieval-augmented generation system using PubMed data and IBM watsonx.ai.
- Integrated biomedical literature retrieval with an interactive React/Three.js frontend.